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ICS 2302 SOFTWARE ENGINEERING 1

This presentation is intended to covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green Exercise tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 7

Requirements Engineering Process and Validation

Learning Outcomes

A study of this lesson should enable students to:

- Describe the requirement engineering process

7.1. Requirements Engineering Process

Requirements engineering is a process that involves all of the activities required to create and maintain a system requirements document. There are four generic, high level requirements engineering process activities. These are a system feasibility study, the elicitation and analysis of requirements, the specification of requirements and their documentation and finally the validation of these requirements. Figure 7.1 illustrates the relationship between these activities. It shows the documents produced at each stage of the requirements engineering (RE) process.

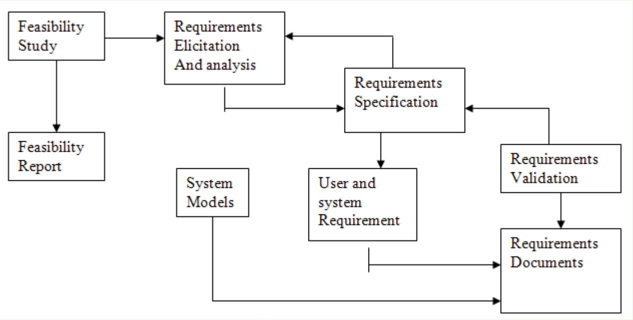

Figure 7.1: Requirements engineering

The requirements engineering activities shown in fig. 7.1are concerned with the elicitation, documentation and checking or requirements.

7.1.1. Feasibility Studies

For all new systems, the requirements engineering process should start with a feasibility study. The input to the feasibility study is an outline description of the system and how it will be used within an organization. The results of the feasibility study should be a report which recommends whether or not it is worth carrying on with the requirements engineering and system development process.

A feasibility study is a short focused study which aims to answer a number of questions:

1. Does the system contribute to the overall objectives of the organization?
2. Can the system be implemented using current technology

and within given cost and schedule constraints?

3. Can the system be implemented with other systems which are already in place?

Carrying out feasibility study involves information assessment information collection and report writing. The information assessment phase identifies the information which is required to answer the three questions set out. Information sources may include the managers of departments where the system will be used, software engineers who are familiar with the type of system that is proposed, technology experts, end-users of the system etc. they should be interviewed during the feasibility study to collect the required information. When the information is available, the feasibility study report is prepared.

This should be a recommendation about whether or not the system development should continue. It may propose changes to the scope, budget and schedule of the system and suggest further high level requirements for the system.

7.1.2. Requirements Elicitation and Analysis

This is the next stage after feasibility studies. In this activity, technical software development staff work with customers and system end-users to find out about the application domain, what services the system provides, the required performance of the system, hardware constraints and so on.

Requirements elicitation and analysis may involve a variety of different kinds of people in an organization. The term stakeholder is used to refer to anyone who should have a direct or indirect influence of the system requirements. Elicitation and analysis is a difficult process for a number of reasons:

1. Stake holders often don't really know what they want from the computer system except in the most general terms; they may find it difficult to articulate what they want from the system; they may make unrealistic demands because they are unaware of the cost of their requests.
2. Stakeholders in a system naturally express requirements in their own terms and with explicit knowledge of their work. Requirements engineers without experience in their customer's domain, must understand the requirements.
3. Different stakeholders have different requirements and they must express these in different ways. Requirement engineers have to discover all potential sources of requirements and discover commonalities and conflict.

4. Political factors may influence the requirements of the system. These may come from managers who demand specific system requirements because these allow them to increase their influence in the organization.

5. The economic and business environment in which the analysis takes place is dynamic. It inevitably changes during the analysis process. Hence the importance of particular requirements may change. New requirements may emerge from new stakeholders who were not originally consulted.

A generic process model of the elicitation and analysis process is shown in figure 7.2 below

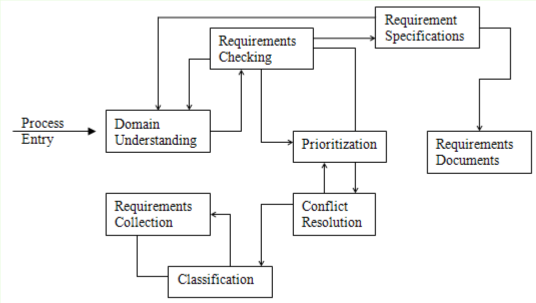

Figure 7.2:

Each organization will have its own version of this general model depending on local factors such as the expertise of the staff, the type of the system being developed, the standards used e.t.c. The process activities are:

1. **Domain understanding** analysts must develop their understanding of the application. For example if it's a system for a supermarket, the analysts must find out how supermarkets operates.

2. **Requirements collection** this is the process of interacting with stake holders in the system to discover their requirements. Obviously, domain understanding develops further during this activity.

3. **Classification** this activity takes the unstructured collection of requirements and organizes them into coherent clusters.

4. **Conflict resolution** inevitably, where multiple stakeholders are involved, requirements will conflict. This activity is concerned with finding and resolving conflicts.

5. **Prioritization** in any set of requirements some will be more important than the others. This stage involves interaction with stakeholders to discover the most important requirements.

6. **Requirements checking** the requirements are checked to discover if they are complete, consistent and in accordance with what stakeholders really want from the system. Figure 7.2shows that elicitation and analysis is an iterative process with continual feedback from each activity to other activities. The process cycle starts with domain understanding and ends with requirement checking. The analysts understanding of the requirement improves with each round of the cycle.

Example ✎. Describe the software requirements document

Solution:

The software requirements document is an agreed statement of the system requirements. It should be organized so that both system customers and software developers can use it. The software requirements document is the official statement of what is required of the system developers.

Should include both a definition of user requirements and a specification of the system requirements.

It is NOT a design document. As far as possible, it should set of WHAT the system should do rather than HOW it should do it. □

7.1.3. Agile methods and requirements

Many agile methods argue that producing a requirements document is a waste of time as requirements change so quickly. The document is therefore always out of date.

Methods such as XP use incremental requirements engineering and express requirements as 'user stories'. This is practical for business systems but problematic for systems that require a lot of pre-delivery analysis (e.g. critical systems) or systems developed by several teams.

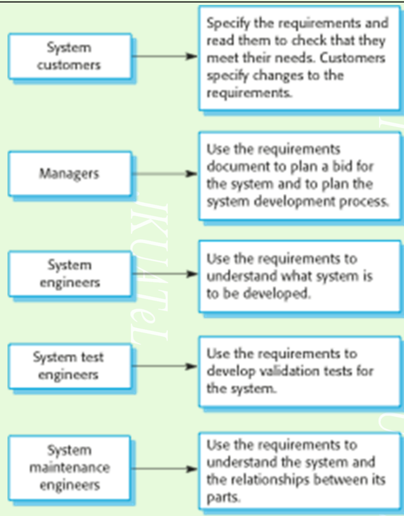

Figure 7.3: The requirements elicitation and analysis process

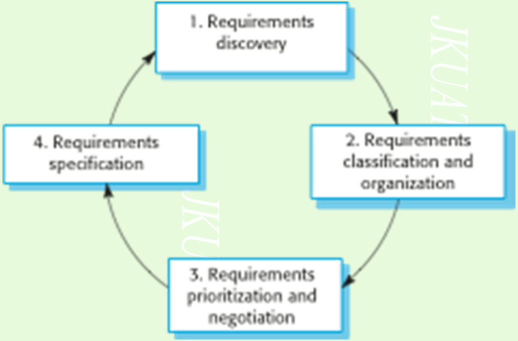

Figure 7.4: Process activities

- Requirements discovery - Interacting with stakeholders to discover their requirements. Domain requirements are also discovered at this stage.

- Requirements classification and organisation - Groups related requirements and organises them into coherent clusters.
- Prioritisation and negotiation - Prioritising requirements and resolving requirements conflicts.
- Requirements specification - Requirements are documented and input into the next round of the spiral.

7.1.4. Problems of requirements elicitation

- Stakeholders don't know what they really want.
- Stakeholders express requirements in their own terms.
- Different stakeholders may have conflicting requirements.

- Organisational and political factors may influence the system requirements.
- The requirements change during the analysis process. New stakeholders may emerge and the business environment change.

7.2. Requirements Validation

Requirements validation is concerned with showing that the requirements actually define the system which the customer wants. It has much in common with the analysis as it is concerned with finding problems with the requirements. However they are distinct processes since validation should be concerned with a complete draft of the requirement document whereas analysis involves working with incomplete requirements. Requirements validation is important because errors in requirements document can lead to extensive rework costs when they are discovered during development or after the system is in service. The cost of making a system change resulting from a requirements problem is much greater than repairing design or coding errors. The reason for this is that a change to the requirements usually means that the system design and implementation must also be changed and that the system must be retested. During the requirements validation process, different types of checks should be carried out on the requirements document. These checks include:

1. Validity checks a user may think that the system is needed to perform certain functions. However, further thought and analysis may identify additional or different functions

that are required. Systems have diverse users with different needs and any set of requirements is inevitably a compromise across the user community.

2. Consistency checks requirements in the document should not conflict. That is, there should not be contradictory constraints or different descriptions of the system function
3. Completeness checks the requirements document should include requirements which define all functions and constraints intended by the system user.
4. Realism checks using knowledge of existing technology, the requirements should be checked to ensure that they can actually be implemented. These checks should also take account of the budget and schedule for the system development.

5. Verifiability to reduce potential for dispute between customer and contractor, system requirements should always be written so that they are verifiable. This means that a set of checks should be designed which can demonstrate that the delivered system meets that requirement.

There are number of requirements validation techniques which can be used in conjunction or individually:-

1. Requirements reviews the requirements are analyzed systematically a team of reviewers.
2. Prototyping in this approach of validation, an executable model of the system is demonstrated to end users and customers. They can experiment with this model to see if it meets their real needs.
3. Test case generation requirements should ideally be testable.

If the tests for the requirements are devised as part of the validation, process, this often reveals requirements problems. If a test is difficult or impossible to design this usually means that the requirements will be difficult to implement and should be reconsidered.

4. Automated consistency analysis if the requirements are expressed as a system model in a structured or formal notation then CASE tools maybe used to check the consistency of the model.

Requirements Management

The requirements for large software systems are always changing. Requirements management is the process of understudying and controlling changes to systems requirements. The process of requirements management is carried out in conjunction with other requirements engineering processes. Planning starts at the same time as the initial requirements elicitation and active management should start as soon as a draft version of the requirements document is available.

Some types of requirements are more susceptible to change than others. This is discussed below:

- Enduring requirement these are relatively stable requirements which derive from the core activity of the organization and which relate directly to the domain of the system. For example, in a hospital there will always be requirements concerned with patients, doctors, nurses, treatment etc.

- Volatile requirements these are requirements which are likely to change during the system development, or after the system has been put into operation. Example of volatile requirements are the requirements resulting from government health care policies.

Requirements management planning

Planning is an essential first stage in the requirement management processes. Requirements management is very expensive and for each project, the planning stage establishes the level of requirements management detail which is required, during this requirements management stage, you have to decide on:

1. Requirements identification each requirements must be uniquely identified so that it can be cross-referenced by other requirements and so that it may be used in traceability as-

sessments

2. A change management process this is the set of activities which assess the impact and cost changes

3. Traceability policies these policies defines the relationships between requirements and between the requirements and the system design that should be recorded and how these records should be maintained.

4. CASE tool support requirements management involves the processing of large amounts of information about the requirements. Tools which maybe used range from specialist requirements management system to spreadsheets and simple database system.

Requirements change management

Requirements change management should be applied to all proposed changes to the requirements. The advantage of using formal process for change management is that all change proposals are treated consistently and that changes to the requirements document are made in a controlled way. There are three principal stages to a change management process:

1. Problem analysis and change specification the process starts with an identified requirements problem or sometimes with a specific change proposal. During this stage, the problem or the change proposals is analyzed to check that it is valid. A more specific requirements change proposal may then be made.

2. Change analysis and costing the effect of the proposed change is assed using traceability information and general knowledge of the systems requirements. The cost of making the change is estimated both in terms of modifications to the requirements document and is appropriate to the system design and implementation. Once this analysis is completed, a decision is made whether or not to proceed with the requirements change.

3. Change implementation the requirement document and where necessary the system design and implementation is modified. The requirements document should be organized so that changes can be accommodated without extensive re-writing.

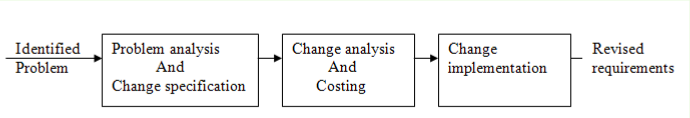

Figure 7.5: Requirements change management

Revision Questions

EXERCISE 1. ✍ Explain the terms requirements engineering process, requirements elicitation and analysis.

References and Additional Reading Materials

1. Hans van Vliet (2005). Software Engineering: Principles and Practice (2nd ed.). John Wiley & Sons, Inc. ISBN: 0-471-97508-7
2. Ian Sommerville, Pete Sawyer(2005). Requirements Engineering: A Good Practice Guide. ISBN: 0-471-97444-7. John Wiley & Sons, Inc
3. Gerald Kotonya, Ian Sommerville (1998). Requirements Engineering: Processes and Techniques. John Wiley &

Sons, Inc. ISBN: 0-471-97208-8

Solutions to Exercises

Exercise 1.

The requirements engineering process is an iterative process including requirements elicitation, specification and validation.

Requirements elicitation and analysis is an iterative process that can be represented as a spiral of activities – requirements discovery, requirements classification and organization, requirements negotiation and requirements documentation.

Exercise 1